# GIS Wetland Degradation & Depletion Analysis

## Multi-Spectral Remote Sensing & Multi-Evidence Change Assessment Report

| | | | |
|---|---|---|---|
| **Run ID:** | run_20260829_114623 | **Baseline (T1):** | 2023-08-14 |
| **Software:** | GIS Engine v1.0.0 | **Present (T2):** | 2023-08-28 |
| **CRS:** | EPSG:32633 | **Interval:** | 14 days |
| **Resolution:** | 10.0 m × 10.0 m | **Analysis Area:** | 36.00 ha |

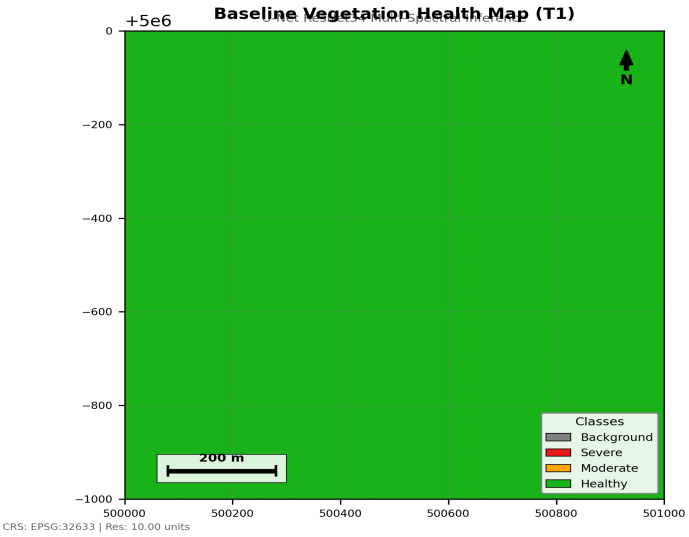
## 1. Executive Summary

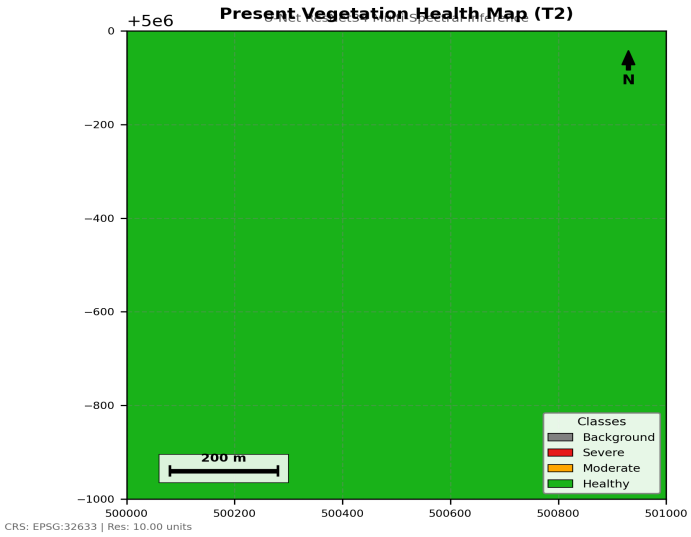
This technical assessment evaluates temporal landscape dynamics between **2023-08-14** (T1) and **2023-08-28** (T2). A total valid overlap footprint of **36.00 ha** was analyzed using a frozen 5-channel ResNet34 U-Net model, 16-state transition analysis, 3D Change Vector Analysis (CVA), and McFeeters NDWI hydrological dynamics. Primary vegetation degradation was detected across **0.00 ha (0.00%)**, while vegetation recovery was observed across **0.00 ha (0.00%)**. The Severity-Weighted Degradation footprint is **0.00 ha** (NDR: **0.0000**). **Physical Wetland Depletion Status:** NOT_ASSESSED.

| Metric | Area (ha) | Percentage (%) | Classification & Scientific Scope |
|---|---|---|---|
| **Vegetation Degradation** | 0.00 | 0.00% | Healthy/Mod → Severe or Healthy → Moderate |
| **Vegetation Recovery** | 0.00 | 0.00% | Severe/Mod → Healthy or Severe → Moderate |
| **Severity-Weighted Footprint** | 0.00 | - | Weighted aggregate: Σ (weight × area) |
| **Surface Water Dynamics** | 0.00 ha | - | NDWI surface water state shift (Loss / Gain) |

## 2. Vegetation Health & Spatial Condition (T1 vs. T2)

Vegetation health was independently inferred for T1 and T2 using the calibrated 5-channel model (NDVI, GNDVI, NDRE, NIR, RedEdge). Four discrete classes are resolved: Background (0), Severe Stress (1), Moderate Stress (2), and Healthy Vegetation (3).

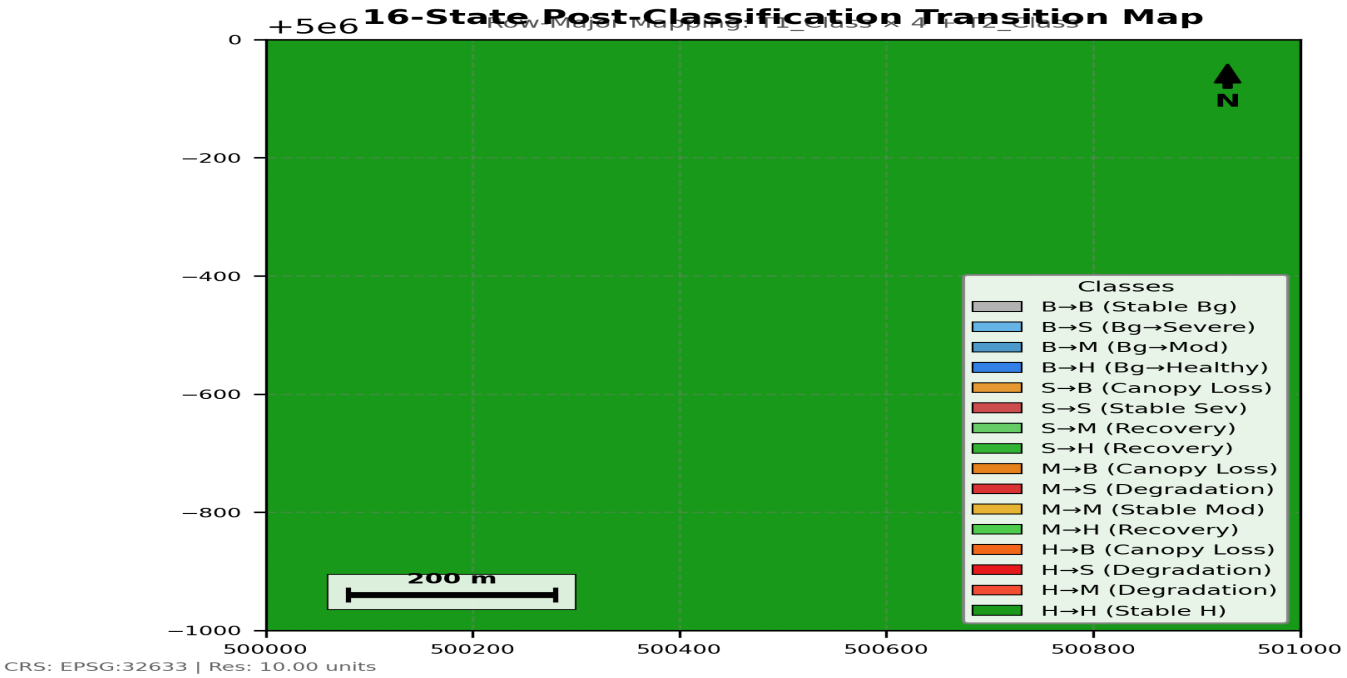



| Vegetation Class | T1 Area (ha) | T1 Pct (%) | T2 Area (ha) | T2 Pct (%) | Net Shift (ha) |
|---|---|---|---|---|---|
| **Healthy Vegetation** | 100.00 | 100.00% | 100.00 | 100.00% | +0.00 |
| **Moderate Stress** | 0.00 | 0.00% | 0.00 | 0.00% | +0.00 |
| **Severe Stress** | 0.00 | 0.00% | 0.00 | 0.00% | +0.00 |
| **Background / Non-Veg** | 0.00 | 0.00% | 0.00 | 0.00% | +0.00 |

## 3. 16-State Post-Classification Transition Analysis

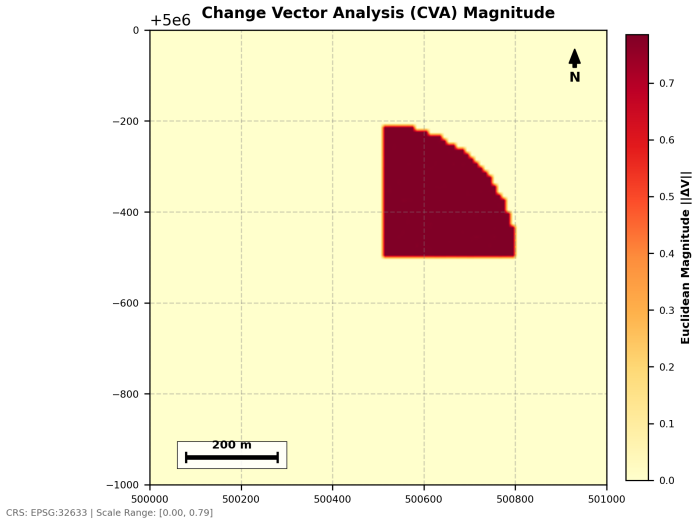
A rigorous cell-by-cell post-classification comparison maps temporal transitions using deterministic row-major indexing (T1_class × 4 + T2_class). This isolates state stability from stress progression, canopy loss, and recovery dynamics.

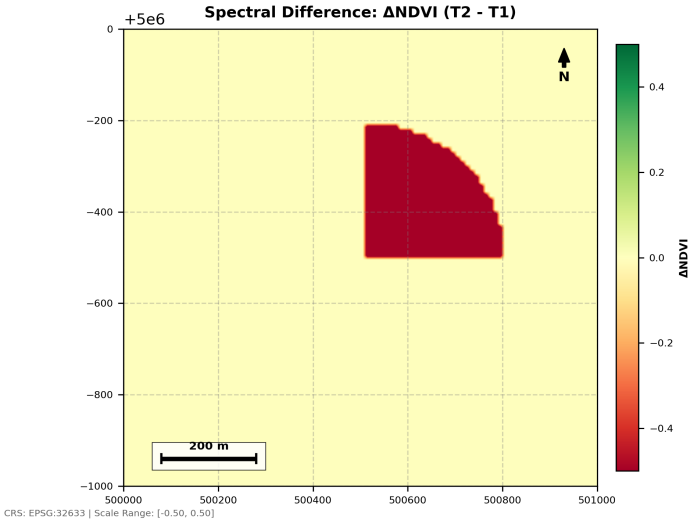



| Transition Category | Transitions Included | Area (ha) | Percentage (%) |
|---|---|---|---|
| **Stable State** | H→H, M→M, S→S, B→B | 100.00 | 100.00% |
| **Vegetation Degradation** | H→M, H→S, M→S | 0.00 | 0.00% |
| **Vegetation Recovery** | M→H, S→M, S→H | 0.00 | 0.00% |
| **Vegetation → Background (Canopy Loss)** | H→B, M→B, S→B | 0.00 | 0.00% |
| **Background → Vegetation (Emergence)** | B→H, B→M, B→S | 0.00 | 0.00% |

## 4. Continuous Change Vector Analysis (CVA) & Residual Stress

Change Vector Analysis evaluates continuous multi-spectral displacement across 3D vector space ([ΔNDVI, ΔGNDVI, ΔNDRE]). CVA magnitude captures subtle intra-class stress shifts without double-counting discrete class transitions.

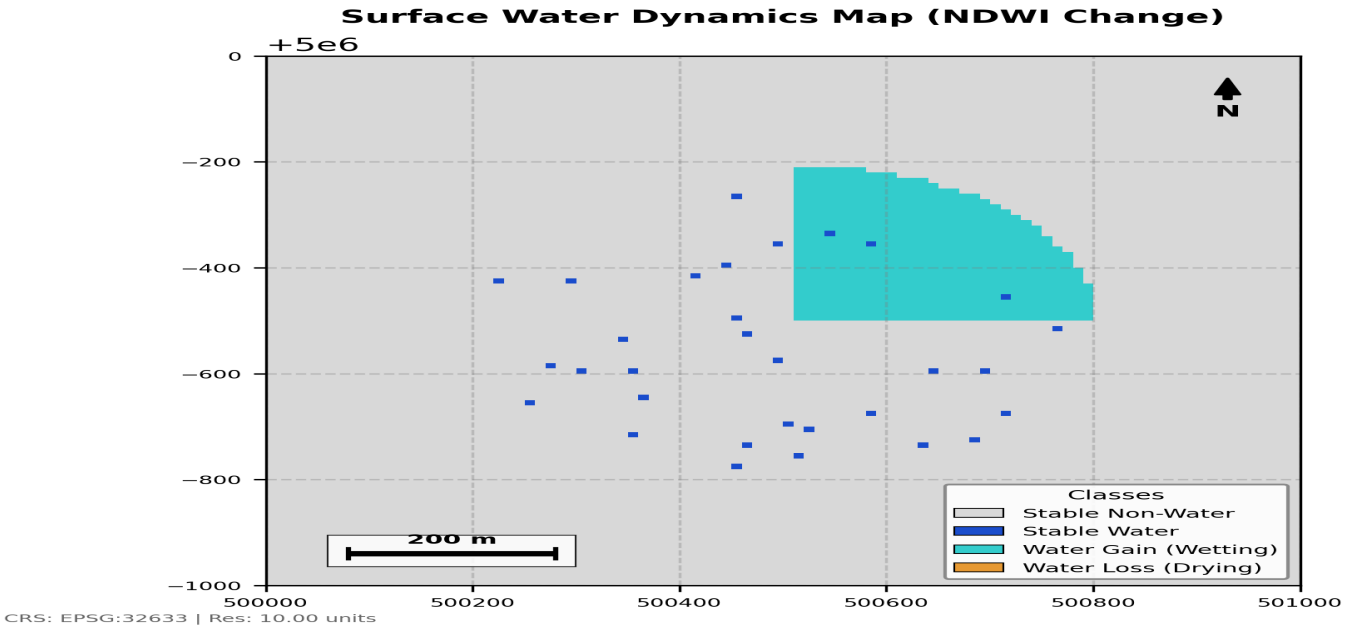


| Spectral Metric | Mean Value | Median Value | 95th Percentile | Physical Interpretation |
|---|---|---|---|---|
| **CVA Magnitude (‖ΔV‖)** | 0.0529 | 0.0000 | 0.7837 | Total spectral vector trajectory displacement |
| **ΔNDVI (T2 - T1)** | -0.0449 | 0.0000 | 0.0000 | Photosynthetic vigor & chlorophyll absorption |
| **ΔGNDVI (T2 - T1)** | -0.0263 | 0.0000 | 0.0000 | Green reflectance & deep canopy density |
| **ΔNDRE (T2 - T1)** | -0.0097 | 0.0000 | 0.0000 | RedEdge boundary & early foliar nitrogen stress |

## 5. Hydrological Evidence Layer (NDWI & Surface Water)

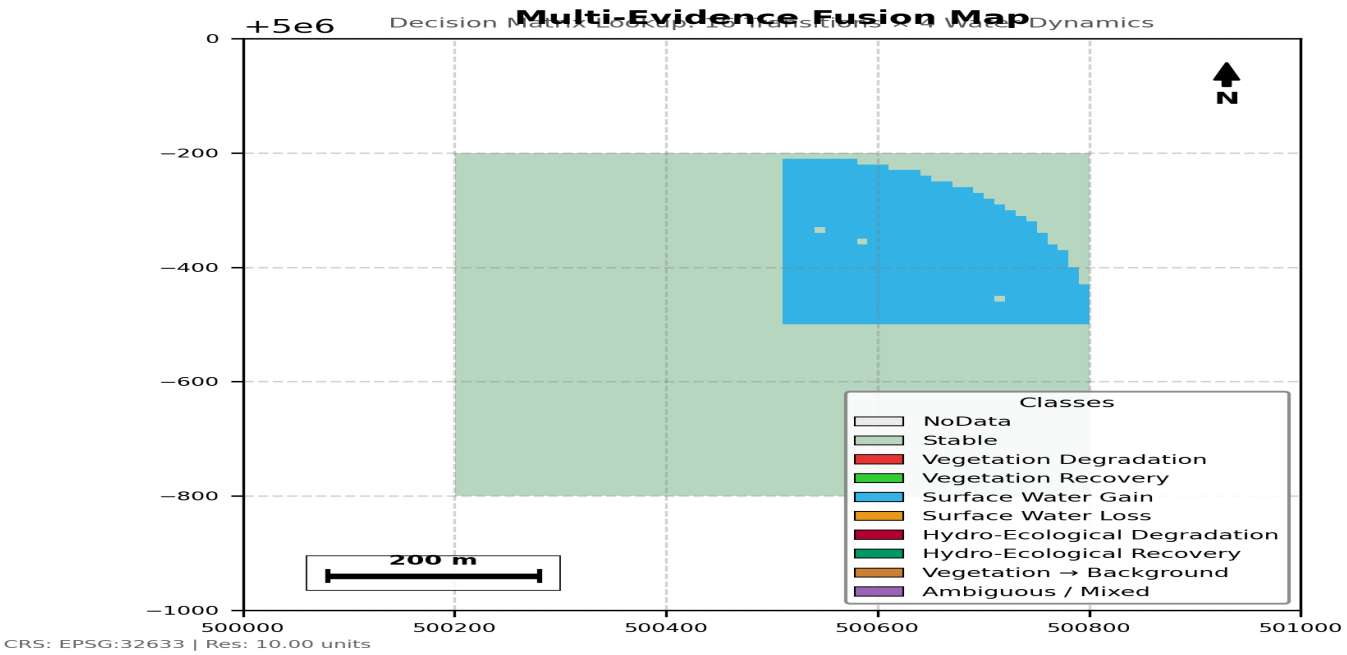
Surface water presence is modeled using the Green/NIR Normalized Difference Water Index (McFeeters 1996: [Green - NIR] / [Green + NIR]). A baseline threshold of 0.0 separates surface water from moist/vegetated soil.




| Hydrological Metric | Area (ha) | Operational Status & Context |
|---|---|---|
| **Baseline Water Extent (T1)** | 0.31 ha | T1 surface water area classified above NDWI threshold (2023-08-14) |
| **Present Water Extent (T2)** | 7.03 ha | T2 surface water area classified above NDWI threshold (2023-08-28) |
| **Surface Water Expansion (Gain)** | 6.72 ha | NDWI shifted below threshold to ≥ 0.0 (Flooding / Wetting) |
| **Surface Water Contraction (Loss)** | 0.00 ha | NDWI shifted from ≥ 0.0 to below threshold (Drying / Receding) |
| **NDWI Water Threshold** | 0.00 | Provisional engineering threshold (Standard McFeeters zero-plane) |

## 6. Multi-Evidence Fusion Matrix

Multi-evidence fusion integrates vegetation health transitions with surface water dynamics using an authoritative 2D decision matrix. This separates true hydro-ecological decline from dryland vegetative stress.




| Fusion Evidence Class | Area (ha) | Multi-Layer Evidence Definition |
|---|---|---|
| **Stable Landscape** | 100.00 ha | Persistent baseline vegetation state and moisture regime |
| **Hydro-Ecological Degradation** | 0.00 ha | Concurrent Vegetation Degradation + Surface Water Loss |
| **Hydro-Ecological Recovery** | 0.00 ha | Concurrent Vegetation Recovery + Surface Water Gain |
| **Vegetation-Only Degradation** | 0.00 | Vegetation health decline occurring over stable water/moisture regime |
| **Vegetation-to-Background (Canopy Loss)** | 0.00 ha | Vegetation loss without independent physical depletion confirmation |

## 7. Quality Assurance, Uncertainty & Production Validation

All analytical steps are checked by the automated Quality Assurance engine to enforce mathematical reconciliation, spatial grid integrity, and conservative scientific bounds.

| Validation Check / Gate | Status | Quality Assurance Finding |
|---|---|---|
| **Input Geometry & CRS Alignment** | PASS | Exact grid match across dimensions, affine transform, and resolution. |
| **Mathematical Area Reconciliation** | PASS | Transition matrix sum matches valid overlap footprint (< 1e-6 ha residual). |
| **Spectral CVA Distribution Check** | PASS | No NaN/Inf leakage; finite bounds verified across all spectral bands. |
| **Independent Ground Reference Data** | UNAVAILABLE | Independent ground-truth samples not configured. Areas are model estimates. |
| **Physical Depletion Claim Guard** | ENFORCED | Safeguard active: Vegetation decline is not labeled physical wetland loss. |

## 8. Scientific Synthesis, Interpretation & Limitations

**OBSERVED (Direct Measurements):** Pixel spectral reflectance changes (ΔNDVI, ΔGNDVI, ΔNDRE), CVA vector magnitude, and NDWI water plane shifts.
**INFERRED (Multi-Layer Evidence):** Vegetation health condition classes, canopy degradation/recovery transitions, and multi-evidence hydro-ecological stress.
**NOT VALIDATED (Uncertainty Notice):** Physical wetland loss/depletion cannot be confirmed without multi-year ground-truth validation or official wetland boundary records.

**Operational Limitations & Technical Notice:**

- **Independent Reference Data:** Ground reference validation samples are currently unavailable. Reported areas represent model-derived estimates rather than error-adjusted Olofsson confidence intervals.
- **Provisional Severity Weights:** Severity weighting factors (Healthy=0.0, Moderate=0.35, Severe=0.80, Background=1.0) are engineering calibration parameters and not field-calibrated ecological constants.
- **Spectral Bandpass Availability:** Drone/Sentinel 4-band data lacks SWIR; hence, MNDWI and LSWI formulations are excluded.
- **Phenological Influence:** Temporal gaps between acquisitions may reflect seasonal senescence rather than permanent anthropogenic degradation.